WEEK 02 REPORT

KNOW YOUR DATA

Project: Moneyball – IPL Player Valuation
Course: 24ADI204 – Data Science and Visualization
Event: Data-Viz Hack-A-Bit (2026–2027)
Team Number: 05

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Submitted as the Week 02 practical documentation for the team project.

1. Executive Summary

Week 02 focused on developing a first impression of the selected IPL dataset. The same merged master dataset used in Week 01 was loaded and inspected using Python libraries. The work examined its dimensions, column names, data types, missing values, duplicate records, unique values, descriptive statistics and season-wise distribution. This inspection establishes the dataset's original condition before the cleaning activities planned for the following week.

Key results: 3,311 records, 27 columns, 17,839 missing cells, zero duplicate records, 794 distinct player names and 19 seasons spanning 2008–2026.

2. Objectives

• Load the project master CSV into a Pandas DataFrame.
• Inspect the dataset structure, dimensions and column data types.
• Identify missing values and calculate their percentages.
• Check duplicate records and unique-value counts.
• Generate numerical and categorical summaries.
• Examine season coverage and player-record frequency.
• Produce a first-impression data quality summary.

3. Dataset Profile

The project uses the merged IPL master dataset named **IPL_Moneyball_Merged_2008_2026.csv**. It contains player-season records and combines performance, auction, team and player-profile information. The dataset is retained as the common source for the weekly project workflow.

Attribute	Observed value
Dataset file	IPL_Moneyball_Merged_2008_2026.csv
Rows / records	3,311
Columns / features	27
Season coverage	2008–2026 (19 seasons)
Distinct player values	794
Duplicate records	0
Total missing cells	17,839

The 27 columns cover season and player identifiers; batting statistics such as runs, balls faced, innings, fours and sixes; bowling statistics such as balls bowled, runs conceded and wickets; auction details; team and player attributes; and derived measures such as strike rate, batting average, economy and bowling average.

4. Tools and Libraries

Python: programming environment used to execute the inspection workflow.
Pandas: loading the CSV, inspecting rows and columns, checking data types, missing values, duplicates, unique values and descriptive summaries.
NumPy: numerical computing support for array-oriented operations.
Matplotlib and Seaborn: visualization libraries used for plots such as the season distribution and missing-value heatmap.

5. Methodology – Work Performed

The Week 02 notebook followed a structured inspection sequence. The CSV was loaded into a DataFrame, and the first, last and randomly selected records were viewed to understand the row structure. The shape and column list were checked to confirm the size and available fields. The info() output and data-type checks were used to distinguish numeric and object columns.

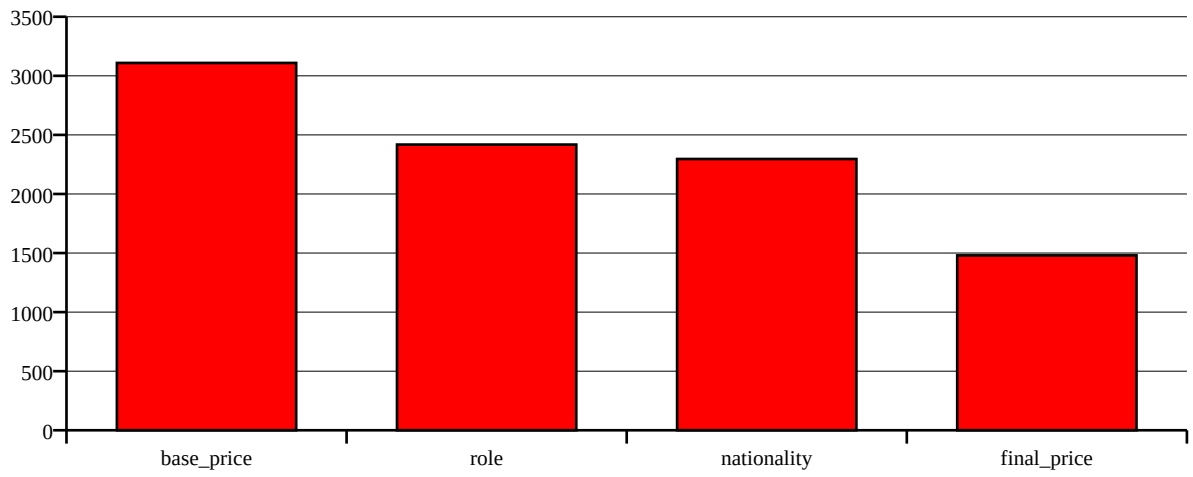
Missing values were counted column-wise and converted into percentages to understand how much information was absent. A heatmap was generated to show the pattern of missingness across the dataset. Duplicate records were checked separately. Unique-value counts were used to understand the variety of values in important fields. Finally, numerical and categorical summaries, season distribution and top-player record counts were inspected.

6. Missing-Value Analysis

The notebook identified **17,839 missing cells** in total. The following columns were among those with substantial missingness:

Column	Missing values	Missing percentage
base_price_lakh	3,109	93.90%
role	2,418	73.03%
nationality	2,296	69.34%
final_price_lakh	1,480	44.70%

Missing values in selected columns



The chart visualizes the exact missing counts listed above. The base_price_lakh field has the highest missing count among these four columns. The missing-value heatmap in the notebook provides a broader visual inspection of where missing entries occur.

Interpretation: Missing values should not automatically be treated as errors. Auction-price fields may be unavailable for players who were not part of a relevant auction record, and some performance fields may be not applicable. The meaning of each field must be checked before choosing an imputation or deletion strategy. Week 02 records the issue; it does not perform cleaning.

7. Duplicate and Unique-Value Analysis

The duplicate-record check returned **0 duplicate rows**. The dataset contains **794 distinct player values** and records across **19 seasons**, from 2008 through 2026. These checks help describe the variety and coverage of the data. A player can appear in more than one season, so the number of distinct players is not expected to equal the number of rows.

8. Descriptive Statistics and Distribution Checks

The notebook generated numerical descriptive statistics to inspect measures such as count, mean, standard deviation, minimum, quartiles and maximum for numeric fields. Categorical summaries were used to inspect the distribution of object-type fields. Season-wise record counts and the ten players with the highest record frequencies were also examined. A season bar chart was generated to make the season distribution easier to compare.

The exact season-wise counts and the names/counts in the top-ten player output are not reproduced here because those individual values are not available in the report-generation record. They remain available in the executed Week 02 notebook and should be retained there as the primary evidence.

9. Data Quality Findings

- **Dataset size:** 3,311 rows and 27 columns, exceeding the project's 1,000-row minimum.
- **Coverage:** 19 seasons from 2008 to 2026.
- **Player variety:** 794 distinct player values.
- **Duplicates:** no duplicate rows were detected.
- **Missingness:** 17,839 missing cells were identified; some columns have high missing percentages.
- **Next step:** review the meaning and pattern of missing values before deciding how to handle them.

10. Before-and-After Status

Week 02 was an inspection and profiling activity. No cleaning, imputation, row deletion, outlier removal, encoding or scaling was performed in this stage. Therefore, there is no processed dataset to compare against the original dataset yet. The Week 02 results form the baseline for the cleaning work planned in Week 03.

11. Conclusion

The Week 02 data inspection provided a clear first impression of the IPL Moneyball master dataset. The dataset contains 3,311 records and 27 columns, covers 19 seasons, includes 794 distinct player values and has no duplicate rows. However, the inspection also revealed 17,839 missing cells, with particularly high missingness in some auction and player-profile fields. These findings identify the areas that require careful investigation during the cleaning stage. The original master dataset is retained unchanged so that later processing decisions can be documented and compared transparently.

12. Project Files and Evidence

The Week 02 GitHub folder should contain exactly these two files:
Week02_Know_Your_Data.ipynb – executed notebook containing the code, outputs and generated visualizations.
Week02_Report.pdf – this report documenting the objectives, methodology, findings and conclusion.

The notebook is the detailed execution evidence. Include or retain the actual Colab screenshots of the dataset inspection, missing-value heatmap, summary outputs and season bar chart in the notebook/report submission as required by the course. This PDF does not claim to contain screenshots that were not available during its preparation.

13. References

1. Project master dataset: IPL_Moneyball_Merged_2008_2026.csv (team-provided merged dataset).
2. Week 02 execution notebook: Week02_Know_Your_Data.ipynb.
3. Data-Viz Hack-A-Bit (2026–2027), Data Science and Visualization, course code 24ADI204 – weekly activity plan

and documentation requirements.

Note: The original three source CSV filenames, source URLs, merge keys and detailed merge procedure were not available in the information used to prepare this report. They have therefore not been invented.